

Some Least Squares Extensions

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Intro

In this lecture, we discuss some extensions of the least squares model.

Specifically, we expand the analysis to incorporate

1. Qualitative information
2. Nonlinear relationships
3. Interactions between explanatory variables

Binary Variables in Regression

Consider the regression model

$$y_i = \mathbf{x}_i' \beta + \gamma d_i + \varepsilon$$

where $d_i = 1$ for some condition occurring and 0 if not. We call these types of variables **dummy variables** or indicator variables.

For example, consider the earnings regression once again, now incorporating a qualitative information:

$$Earnings = \beta_1 + \beta_2 Educ + \beta_3 Age + \beta_4 union + \varepsilon$$

where $union = 1$ if the worker is registered in a union and 0 otherwise.

How do we interpret β_4 ?

Binary Variables in Regression

Dummy variables are particularly useful in loglinear regressions.

Consider the model

$$\ln y = \beta_1 + \beta_2 x + \beta_3 d + \varepsilon$$

The coefficient d indicates a multiplicative shift of the function.

Binary Variables in Regression

The percentage change in $E[\mathbf{y}|\mathbf{X}, d]$ associated with the change in d is

$$\begin{aligned}\%(\Delta E[\mathbf{y}|\mathbf{X}, d]/\Delta d) &= 100\% \left(\frac{E[\mathbf{y}|\mathbf{X}, d = 1] - E[\mathbf{y}|\mathbf{X}, d = 0]}{E[\mathbf{y}|\mathbf{X}, d = 0]} \right) \\ &= 100\%[\exp(\beta_3) - 1]\end{aligned}$$

Take the Monet paintings example again. We can introduce qualitative information for a better model fit.

On the data set, we also have information whether the painting was signed by Monet or not (of the 430 sales in the sample, 77 are for unsigned paintings). This definitely affects its price!

Monet's Paintings

The table below shows the regression results from the model

$$\ln(\text{Price}) = \beta_1 + \beta_2 \ln(\text{Area}) + \beta_3 \text{Aspect Ratio} + \beta_4 \text{Signature} + \varepsilon$$

TABLE 6.2 Estimated Equations for Log Price

$$\ln \text{price} = \beta_1 + \beta_2 \ln \text{Area} + \beta_3 \text{Aspect Ratio} + \beta_4 \text{Signature} + \varepsilon$$

Mean of $\ln \text{Price}$ 0.33274

Number of observations 430

Sum of squared residuals	520.765	420.609
Standard error	1.10435	1.35024
R-squared	0.33417	0.46223
Adjusted R-squared	0.33105	0.45844

<i>Variable</i>	<i>Coefficient</i>	<i>Standard Error</i>	<i>t Ratio</i>	<i>Coefficient</i>	<i>Standard Error</i>	<i>t Ratio</i>
<i>Constant</i>	-8.34327	0.67820	-12.30	-9.65443	0.62397	-15.47
<i>ln Area</i>	1.31638	0.09205	14.30	1.34379	0.08787	16.22
<i>Aspect ratio</i>	-0.09623	0.15784	-0.61	-0.01966	0.14222	-0.14
<i>Signature</i>	—	—	—	1.26090	0.12519	10.07

Monet's Paintings

Computing this result for a painting of the same area and aspect ratio, we find the model predicts that the signature effect would be

$$100\% \times \frac{\Delta E(Price)}{Price} = 100\%[\exp(1.26) - 1] = 256\%$$

The effect of a signature on an otherwise similar painting is to more than double the price.

Using the delta method, we obtain an estimated standard error for $[\exp(b_3)^2 - 1]$ of the square root of $[\exp(b_3)] \times 0.1253^2$, which is 0.4417.

For the percentage difference of 252%, we have an estimated standard error of 44.17%.

Several Categories

When there are several categories, a set of binary variables is necessary.

Consider a consumption function for quarterly data:

$$C_t = \beta_1 + \beta_2 x_t + \delta_1 D_{1t} + \delta_2 D_{2t} + \delta_3 D_{3t} + \varepsilon_t$$

Notice that we only introduce 3 quarterly dummy variables, as the fourth one would add one to all observations, adding to the intercept. This is known as the **dummy variable trap**.

Several Categories

The preceding is a means of **deseasonalizing** the data. Consider the alternative formulation:

$$C_t = \beta x_t + \delta_1 D_{1t} + \delta_2 D_{2t} + \delta_3 D_{3t} + +\delta_4 D_{4t} + \varepsilon_t$$

The FWL theorem suggests that we can obtain β by regressing both C and x on the four quarterly dummies and then regress the residuals of the first on the residuals of the second.

This means we are *removing* the variation in both C and x associated with each quarter of the year.

Threshold Effects and Categorical Variables

Suppose we are interested in estimating the effect of education on earnings and the available data consists of the highest level of educational attainment, such as less than high school (LTHS), high school (HS), college (C), post-graduate (PG).

A flexible model would use three (or four) binary variables, one for each level of education:

$$Earnings = \beta_1 + \beta_2 Age + \delta_{HS} HS + \delta_C C + \delta_{PG} PG + \varepsilon$$

How do we interpret these coefficients?

Threshold Effects and Categorical Variables

The correspondence between the coefficients and income for a given age is

$$\text{LTHS: } E[\text{Earnings}|\text{Age}, \text{LTHS}] = \beta_1 + \beta_2 \text{Age}$$

$$\text{HS: } E[\text{Earnings}|\text{Age}, \text{HS}] = \beta_1 + \beta_2 \text{Age} + \delta_{\text{HS}}$$

$$\text{C: } E[\text{Earnings}|\text{Age}, \text{C}] = \beta_1 + \beta_2 \text{Age} + \delta_{\text{C}}$$

$$\text{PG: } E[\text{Earnings}|\text{Age}, \text{PG}] = \beta_1 + \beta_2 \text{Age} + \delta_{\text{PG}}$$

We can then estimate, for example, $\delta_{\text{PG}} - \delta_{\text{C}}$, which might be of interest.

Threshold Effects and Categorical Variables

Another way to estimate the relationship above is to recast the variables in terms of whether the individual has some degree and not the highest level attained.

How would you redefine the above equations? How would we interpret each dummy variable coefficient?

Transition Tables

Hodge and Shankar (2014) propose a useful framework for arranging the effects of a sequence of categories:

TABLE 6.7 Education Effects in Estimated Log Wage Equation

Effects of switches between categories in education level

<i>Initial Education</i>	<i>New Education</i>	<i>Partial Effect</i>	<i>Standard Error</i>	<i>t Ratio</i>
<i>LTHS</i>	<i>HS</i>	0.13832	0.03351	4.13
<i>LTHS</i>	<i>COLLEGE</i>	0.29168	0.04181	6.98
<i>LTHS</i>	<i>POSTGRAD</i>	0.40651	0.04896	8.30
<i>HS</i>	<i>COLLEGE</i>	0.15336	0.03047	5.03
<i>HS</i>	<i>POSTGRAD</i>	0.26819	0.03875	6.92
<i>COLLEGE</i>	<i>POSTGRAD</i>	0.11483	0.03787	3.03

How would we calculate the standard error of the partial effects?

Nonlinearities in the Variables

Let $\mathbf{z} = z_1, z_2, \dots, z_L$ be a set of L independent variables; let $f_1, f_2, \dots, f_K$ be K linearly independent functions of $\mathbf{z}$; let $g(y)$ be an observable function of y ;

The linear regression model may be written

$$\begin{aligned} g(y) &= \beta_1 f_1(\mathbf{z}) + \beta_2 f_2(\mathbf{z}) + \dots + \beta_K f_K(\mathbf{z}) + \varepsilon \\ &= \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_K x_K + \varepsilon \\ &= \mathbf{x}'\beta + \varepsilon \end{aligned}$$

This linear model can be tailored to any number of situations.

Functional Forms

A commonly used form of regression model is the loglinear model,

$$\ln y = \ln \alpha + \sum_k \beta_k \ln x_k + \varepsilon = \beta_1 + \sum_k \beta_k \ln x_k + \varepsilon$$

In this model, the coefficients are elasticities:

$$\left(\frac{\partial y}{\partial X_k} \right) \left(\frac{X_k}{y} \right) = \frac{\partial \ln y}{\partial \ln X_k} = \beta_k$$

A **semilog** model is

$$\ln y = \beta_1 + \beta_2 x + \varepsilon$$

How do we interpret β_2 ?

Functional Forms

An alternative approach sometimes taken is to measure the variables and associated changes in standard deviation units:

$$x_{ik}^* = (x_{ik} - \bar{x}_k)/s_k, y_i^* = (y_i - \bar{y})/s_y$$

How do we interpret the β_2 coefficients in a regression model given by

$$y_i^* = \beta_1 + \beta_2 x_i^* + \varepsilon$$

Interaction Effects

Another useful formulation of the regression model is one with interaction terms.

Suppose the following model:

$$\ln wage = \beta_1 Educ + \beta_2 female + \beta_3 Educ \times female + \cdots + \varepsilon$$

In this model,

$$\frac{\partial E[\ln wage | Educ, fem, \cdots]}{\partial Educ} = \beta_1 + \beta_3 female$$

which implies that the marginal effect of education differs between men and women.

Interaction Effects

To perform hypothesis testing and constructing confidence intervals, the necessary standard error is computed from the square root of

$$\begin{aligned} Var \left(\frac{\partial E[\ln wage | E\hat{duc}, fem, \dots]}{\partial E\hat{duc}} \right) = & Var[\hat{\beta}_1] + female^2 Var[\hat{\beta}_3] \\ & + 2female Cov(\hat{\beta}_1, \hat{\beta}_3) \end{aligned}$$

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